

3. (a) Crude oil can be separated into simpler mixtures called fractions. These fractions contain hydrocarbon compounds called alkanes. **Table 1** shows information about some of the fractions obtained from crude oil by fractional distillation.

Fraction	Boiling point range (°C)	Number of carbon atoms present in the alkanes
petroleum gases	< 20	C ₁ -C ₄
petrol	30-75	C ₅ -C ₁₀
naphtha	70-170	C ₈ -C ₁₂
kerosene	170-250	C ₁₀ -C ₁₄
diesel oil	250-340	C ₁₄ -C ₂₄
lubricating oil	340-500	C ₂₁ -C ₃₀
fuel oil	490-580	C ₂₅ -C ₃₅
residue	> 580	>C ₃₅

Table 1

Use **only** the information in **Table 1** to answer parts (i)-(iii).

- (i) Hexane has a boiling point of 68 °C. Give the name of the fraction which contains hexane. [1]

.....

- (ii) One alkane is found in kerosene and in diesel. Give the number of carbon atoms in this alkane. [1]

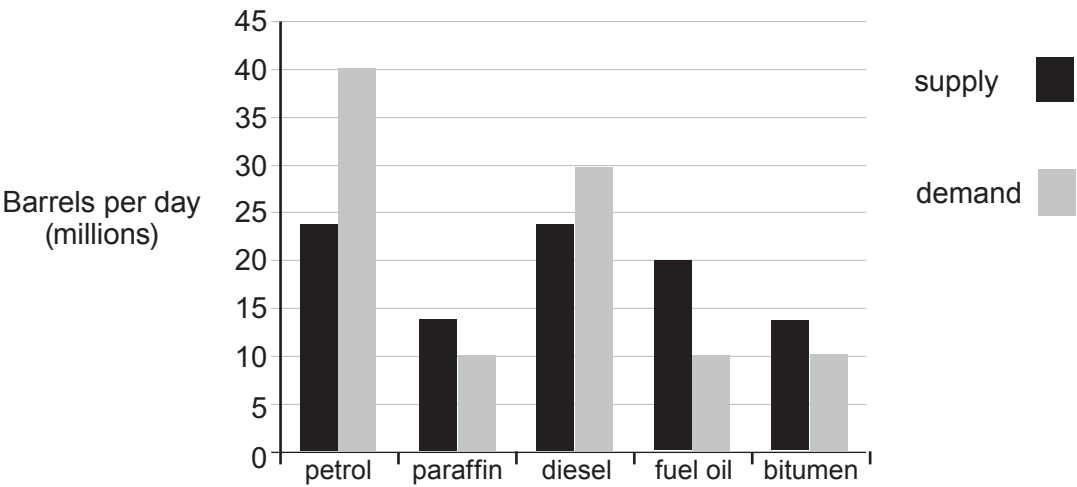
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- (iii) Give the number of carbon atoms in the alkane which has the **lowest** boiling point. [1]

.....



(b) The bar chart shows the **supply** and **demand** for some crude oil products.



Name the fractions for which the demand is greater than the supply. Suggest a reason why these fractions are in high demand. [2]

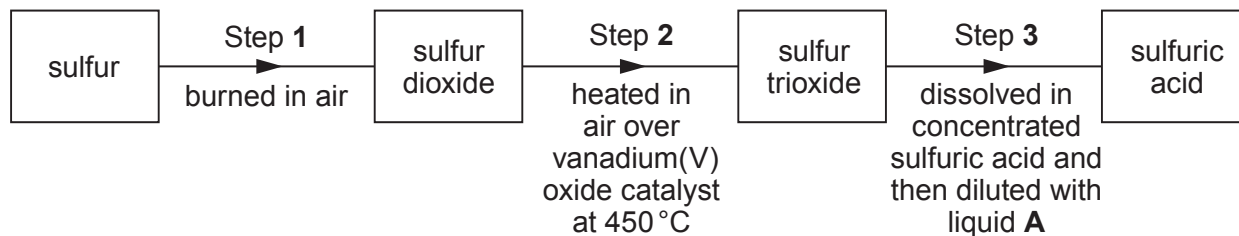
Fractions and

Reason

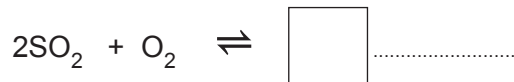
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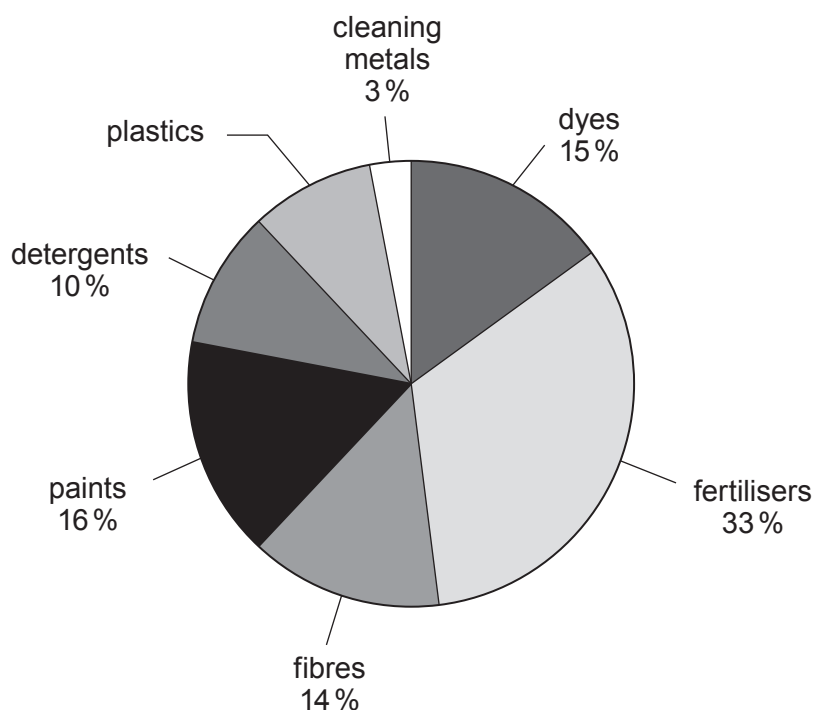
3. (a) The flow diagram shows steps in the manufacture of sulfuric acid using the Contact Process.



- (i) Name the **element** in the air which reacts with sulfur in step 1. [1]
.....
- (ii) State the purpose of the catalyst in step 2. [1]
.....
- (iii) Name liquid **A** used to dilute the concentrated sulfuric acid in step 3. [1]
.....
- (iv) Complete and balance the symbol equation for the reaction in step 2. [2]



(b) The pie chart shows the main uses of sulfuric acid.



Calculate the percentage of sulfuric acid used for making plastics.

[2]

Percentage = %

(c) One important use of sulfuric acid is in the production of the fertiliser ammonium sulfate, $(\text{NH}_4)_2\text{SO}_4$.

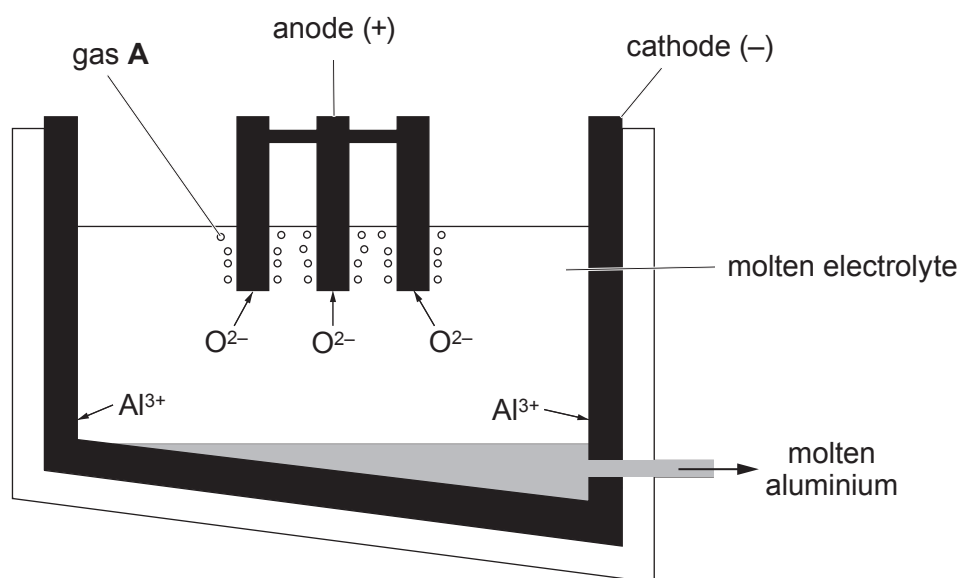
Complete the following word equation for the production of ammonium sulfate.

[1]

sulfuric acid + $\longrightarrow$ ammonium sulfate



3. (a) The diagram shows a cell used in the extraction of aluminium from aluminium oxide.



positive	oxygen	cryolite	negative
aluminium	hydrogen	steel	graphite

- (i) Choose words from the box to complete each of the following sentences. [4]

The electrolyte is a mixture of molten aluminium oxide and

Aluminium ions, Al^{3+} , are attracted to the electrode with a charge.

Gas **A** is

The electrodes are made of

- (ii) Balance the equation for the overall reaction. [1]



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(b) Calculate the relative formula mass (M_r) of aluminium oxide, Al_2O_3 . [2]

$A_r(\text{O}) = 16$ $A_r(\text{Al}) = 27$

Relative formula mass =

(c) The box below contains some properties of aluminium.

low density	non-magnetic	resists corrosion
good thermal conductor	good electrical conductor	

Window frames can be made from different materials.

Choose **two** properties from the box which make aluminium a **better** material than iron for making window frames. [2]

Property 1

Property 2

9

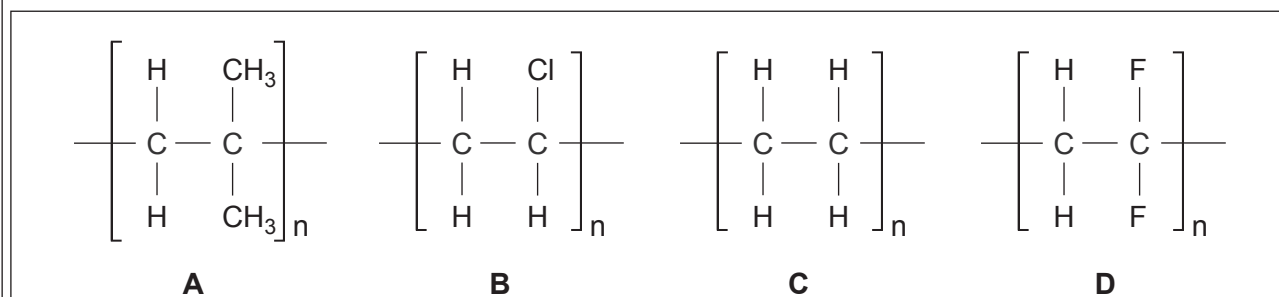
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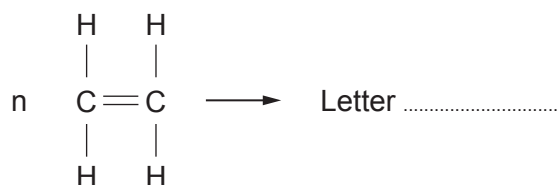
3. (a) Complete the table.

[2]

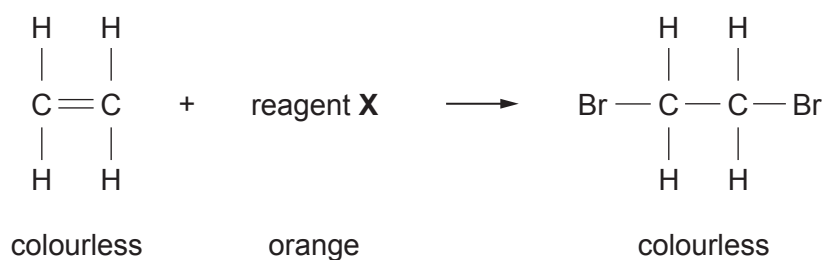
Name	Molecular formula	Structural formula
ethane	C_2H_6	<pre> H H H — C — C — H H H </pre>
propane	C_3H_8	
butane		<pre> H H H H H — C — C — C — C — H H H H H </pre>

(b) The box contains four polymers **A**, **B**, **C** and **D**.Give the **letter** of the polymer formed during the polymerisation of ethene.

[1]



(c) The equation shows the reaction taking place when testing for a $C=C$ bond.



bromine water

potassium dichromate

universal indicator

barium chloride

Choose the name of reagent **X** from the box above.

.....

[1]

(d) The equation shows the fermentation of glucose.



(i) Underline the chemical name for $\text{C}_2\text{H}_5\text{OH}$.

[1]

sugar

biofuel

ethanol

alcohol

(ii) Calculate the relative molecular mass, M_r , of $\text{C}_2\text{H}_5\text{OH}$.

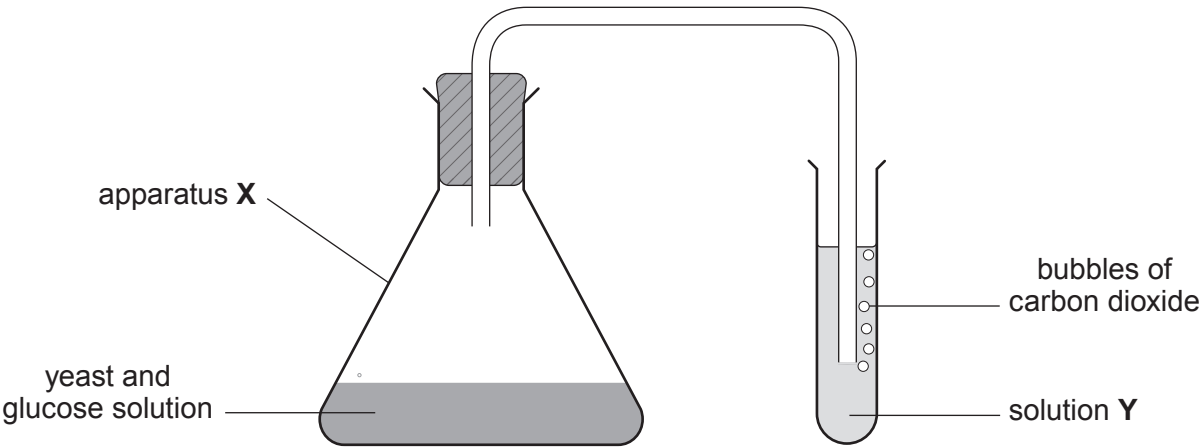
[2]

$$A_r(\text{H}) = 1 \quad A_r(\text{O}) = 16 \quad A_r(\text{C}) = 12$$

$$M_r = \dots\dots\dots$$



3. (a) Ethanol can be produced by fermentation using the apparatus shown.



(i) Circle the name of apparatus X. [1]

conical flask measuring cylinder beaker

(ii) Solution Y can be used to show the presence of carbon dioxide gas. [2]

Give the name of solution Y and state what is seen when carbon dioxide is bubbled through it.

Name of solution Y

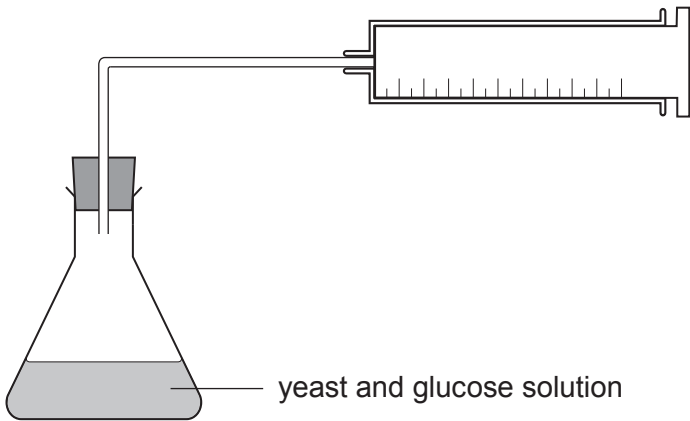
What is seen

(iii) The chosen temperature for this reaction is 35 °C. Tick (✓) the correct reason why carbon dioxide would **not** be produced at a temperature of 90 °C. [1]

the reaction is finished	
the yeast is used up	
the enzymes in the yeast are denatured	



(b) In a different experiment, using the apparatus shown below, the volume of carbon dioxide produced at 35 °C was measured and recorded every 10 minutes for 60 minutes.



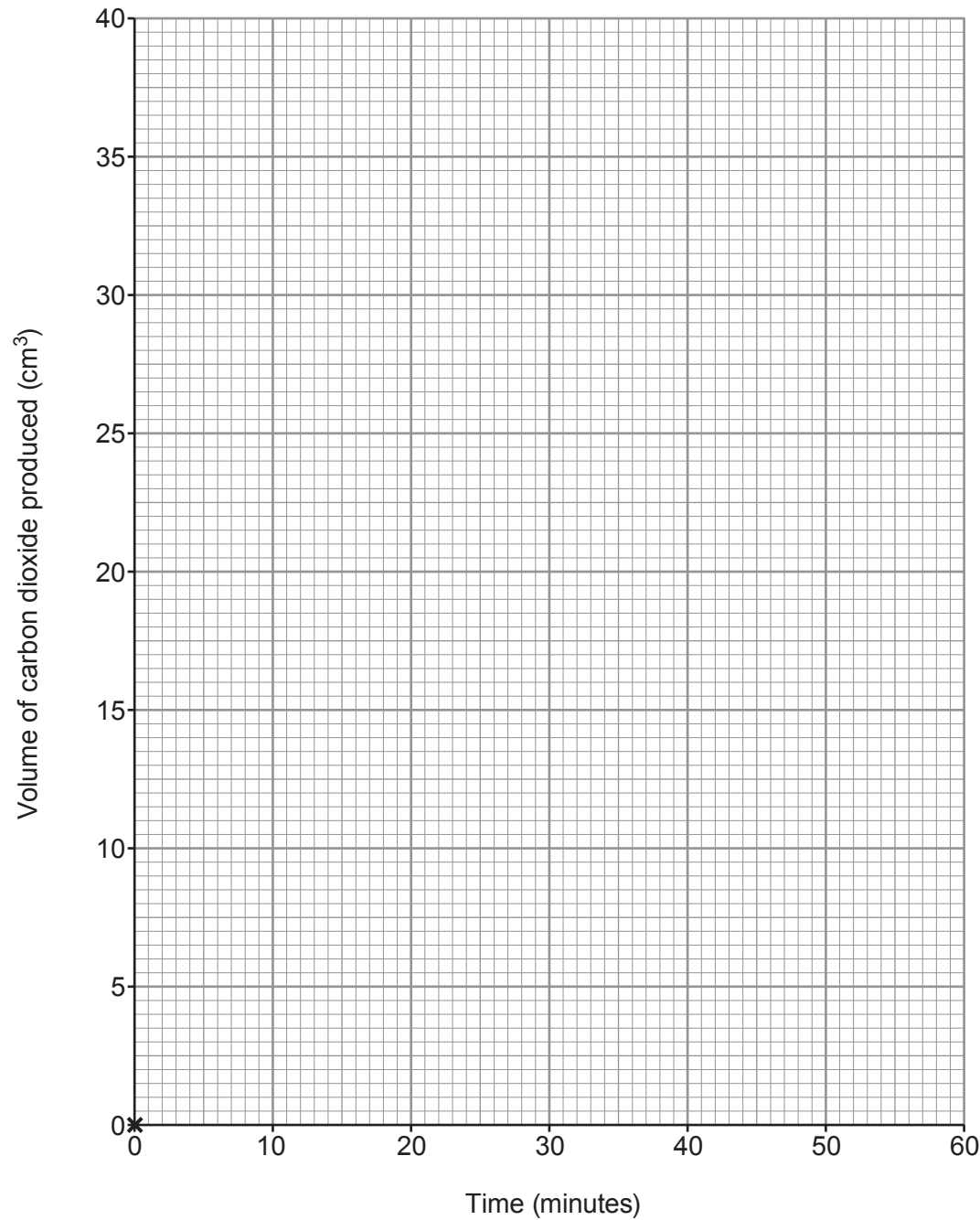
Time (minutes)	Volume of carbon dioxide produced (cm ³)
0	0
10	6
20	12
30	17
40	25
50	30
60	36



(i) Plot the data on the grid below and draw a suitable line.

The first point has been plotted for you.

[3]



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(ii) Use the graph to find the volume of carbon dioxide produced after 25 minutes. [1]

Volume cm³

(iii) Use the data to **estimate** how long it would take to produce 100 cm³ of carbon dioxide gas. Assume that the rate of the reaction does not change. [1]

Time = minutes

9

